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public static String toString(int[] a)

Arrays class has provided a predefined static method called toString() through which we
can convert the array elements into String format.

It can accept(int []arr) as a parameter and return type is String.

We have various overloaded methods

package com.ravi.one_diemensional_basic;

import java.util.Arrays;

/*Q1) Different ways to create 1D array and fetching the elements using
ordinary for loop /forEach loop and toString().*/

public class ArrayDemo1
{
    public static void main(String[] args)
    {
        int []arr1 = new int[3];
        arr1[0] = 10;
        arr1[1] = 20;
        arr1[2] = 30;

        IO.println("Retrieving the array elements using ordinary for loop :");

        for(int i=0; i<arr1.length; i++)
        {
            IO.println(arr1[i]);
        }

        int []arr2 = new int[] {100,200,300};
        IO.println("Retrieving the array elements using for each loop :");

        for(int element : arr2)
        {
            IO.println(element);
        }

        IO.println("Printing the array elements using toString() :");

        int []arr3 = {1000, 2000, 3000};
        IO.println(Arrays.toString(arr3));
    }
}

package com.ravi.one_diemensional_basic;

//Q2) WAP to print the array elements in reverse order. (user input)
//[Array size must be a positive Integer only]

public class ArrayDemo2
{
    public static void main(String[] args)
    {
        int size = Integer.parseInt(IO.readLine("Enter the size of the array :"));

        int []arr = new int[size];

        //Initializing the array elements

        for(int i=0; i<size; i++)
        {
            IO.print("Enter the element at "+i+" index ");
            arr[i] = Integer.parseInt(IO.readLine());
        }

        IO.println("Printing the Array elements :");
        for(int i=0; i<arr.length; i++)
        {
            IO.println(arr[i]);
        }

        IO.println("Printing the Array elements in reverse order :");

        //10 20 30 40 50 //length = 5

        for(int i=arr.length-1; i>=0; i--)
        {
            IO.println(arr[i]);
        }
    }
}

package com.ravi.one_diemensional_basic;

import java.util.Arrays;

//Q3) WAP to find the the sum of all elements of the array.

class FindSum
{
    public static int getSumOfAllTheElements(int []arr)
    {
        int sum = 0;

        for(int element : arr)
        {
            sum += element;
        }

        return sum;
    }
}

public class ArrayDemo3
{
    public static void main(String[] args)
    {
        int []val = {100,200,300,400,500};
        IO.println("Original Array :"+Arrays.toString(val));

        int sum = FindSum.getSumOfAllTheElements(val);
        IO.println("Sum of all the elements are :"+sum);
    }
}

package com.ravi.one_diemensional_basic;

//WAP to calculate the student marks and find out the total average.

public class ArrayDemo4
{
    void main()
    {
        int noOfSubjects = Integer.parseInt(IO.readLine("Enter number of subjects :"));

        int []marks = new int[noOfSubjects];

        //Initialize the Array
        for(int i=0; i<marks.length; i++)
        {
            IO.print("Enter "+(i+1)+" subject marks :");
            marks[i] = Integer.parseInt(IO.readLine());
        }

        //Finding the total marks

        double total = 0.0;

        for(int mark : marks)
        {
            total += mark;
        }

        IO.println("Total Marks is :"+total);
        IO.println("Average is :"+(total/noOfSubjects));
    }
}

package com.ravi.one_diemensional_basic;

//WAP to find the minimum element in the Array

public class ArrayDemo5 {

    public static void main(String[] args)
    {
        int size = Integer.parseInt(IO.readLine("Enter the size of the array :"));

        int []arr = new int[size];

        for(int i=0; i<size; i++)
        {
            IO.print("Enter the element at "+i+" index :");
            arr[i] = Integer.parseInt(IO.readLine());
        }

        int min = arr[0];          //9  5  8  1              min = 1

        for(int element : arr)
        {
            if(element < min)      //1 < 5
            {
                min = element;
            }
        }
        IO.println("Min element in the array is :"+min);
    }
}

package com.ravi.one_diemensional_basic;

//Q6) WAP to work with reference array :
//a) Manual Insertion

record Customer(Integer id, String name, Double bill)
{
}

public class ArrayDemo6 {

    public static void main(String[] args)
    {
        Customer customers[] = new Customer[3];

        customers[0] = new Customer(111, "Scott", 120000);
        customers[1] = new Customer(222, "Smith", 150000);
        customers[2] = new Customer(333, "John", 180000);

        for(Customer customer : customers)
        {
            IO.println(customer);
        }
    }
}

package com.ravi.one_diemensional_basic;

//WAP to store the Product object in the array and search the product
//object based on the id

record Product(Integer id, String name, Double price)
{
}

public class ArrayDemo7
{
    public static void main(String[] args)
    {
        Integer size = Integer.valueOf(IO.readLine("Enter the size of the array :"));

        Product []products = new Product[size];

        //Initialize the array
        for(int i=0; i<size; i++)
        {
            Integer id = Integer.valueOf(IO.readLine("Enter product id :"));
            String name = IO.readLine("Enter Product Name :");
            Double price = Double.valueOf(IO.readLine("Enter product Price :"));

            products[i] = new Product(id, name, price);
        }

        IO.println("Printing the Original Array :");

        for(Product product : products)
        {
            IO.println(product);
        }

        Integer id = Integer.valueOf(IO.readLine("Enter the id to search a product :"));
        boolean isFound = false;

        for(Product product : products)
        {
            if(product.id().equals(id))
            {
                isFound = true;
                IO.println("Product with id "+id+" is available");
                IO.println(product);
                break;
            }
        }

        if(!isFound)
        {
            System.err.println("Product with id "+id+ " is not available");
        }
    }
}

0      1      2
null    null    null

package com.ravi.one_diemensional_basic;

//Initializing the object properties using index

class Book
{
    String title;
    String author;
}

public class ArrayDemo8 {

    public static void main(String[] args)
    {
        Book books[] = new Book[3];
        books[0] = new Book();
        books[1] = new Book();
        books[2] = new Book();

        books[0].title = "Java";
        books[0].author = "Jama";

        books[1].title = "Python";
        books[1].author = "Scott";

        books[2].title = "HTML";
        books[2].author = "Lee";

        int index = 0;

        while(index < books.length)
        {
            IO.print(books[index].title);
            IO.print(" By ");
            IO.print(books[index].author);
            index++;
            IO.println();
        }
    }
}

package com.ravi.one_diemensional_basic;

import java.util.Arrays;

//Q8) WAP to modify the value of an array.[Change Array Element]

class ModifyArrayElement
{
    public static int[] changeArrayElement(int []arr) //arr - 10, 20, 30 ,40,50
    {
        arr[0] = 100;
        arr[1] = 200;

        return arr;
    }
}

public class ArrayDemo9
{
    public static void main(String[] args)
    {
        int []values = {10,20,30,40,50};
        IO.println("Original Array :"+Arrays.toString(values));

        int[] modifiedArray = ModifyArrayElement.changeArrayElement(values);
        IO.println("Modified Array :"+Arrays.toString(modifiedArray));

        IO.println("After modification Original Array :"+Arrays.toString(values));
    }
}

values
arr
modifiedArray
0      1      2      3      4
10     20     30     40     50
100    200

[Array object is
created at 1000x
memory loaction]
```